

CS & IT ENGINEERING



Computer Network

Transport Layer

Lecture No. - 13

By - Abhishek Sir





Recap of Previous Lecture



Topic

Socket Programming



Topics to be Covered



Topic

TCP Options

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW



#Q. A client process P needs to make a TCP connection to a server process S. Consider the following situation: the server process S executes a `socket()`, a `bind()` and a `listen()` system call in that order, following which it is preempted. Subsequently, the client process P executes a `socket()` system call followed by `connect()` system call to connect to the server process S. The server process has not executed any `accept()` system call. Which one of the following events could take place?

[GATE-2008]

115C

~~(A)~~ `connect()` system call returns successfully FALSE

~~(B)~~ `connect()` system call blocks

✓ (C) `connect()` system call returns an error

~~(D)~~ `connect()` system call results in a core dump

Ans: C



Topic : Socket Programming with TCP



IP

TCP Client

① socket()

② connect()

TO

Error

S

TCP Server

socket() ①

bind() ②

listen() ③

SYN



#Q. Consider socket API on a Linux machine that supports connected UDP sockets.
A connected UDP socket is a UDP socket on which connect function has
already been called. Which of the following statements is/are CORRECT?

I. A connected UDP socket can be used to communicate with multiple peers
simultaneously. FALSE

II. A process can successfully call connect function again for an already
connected UDP socket. TRUE

[GATE-2017, Set-2]

IIT-R

(A) I only

✓ (B) II only

(C) ~~I~~ Both I and II

(D) Neither I nor II

Ans: B

#Q. Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let α be the value of RTT in milliseconds (rounded off to the nearest integer) after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are.

[GATE-2015, Set-2, 2-Mark]

~~(A)~~ 63 milliseconds, 65535×2^{14}

~~(B)~~ 63 milliseconds, 65535×2^{16}

☒ (C) 500 milliseconds, 65535×2^{14}

~~(D)~~ 500 milliseconds, 65535×2^{16}

$$\alpha = \frac{(2^{16} - 1) \text{ bytes}}{\text{Bandwidth}} = \frac{(2^{16} - 1) * 2^3 \text{ bits}}{1048560 \text{ bits/sec}}$$

$$\alpha = 0.5 \text{ sec} = 500 \text{ ms}$$

$$\beta = [(2^{16} - 1) * \text{Scale_factor}] \text{ bytes}$$

$$= [(2^{16} - 1) * 2^{\text{shift_count}}] \text{ bytes}$$

$$= (2^{16} - 1) * 2^{14} \text{ bytes}$$

Ans: C

#Q. Consider the following statements about the timeout value used in TCP.

i. The timeout value is set to the RTT (Round Trip Time) measured during TCP connection establishment for the entire duration of the connection. FALSE

ii. Appropriate RTT estimation algorithm is used to set the timeout value of a TCP connection. TRUE

iii. Timeout value is set to twice the propagation delay from the sender to the receiver. FALSE

$$RTT > 2 \times p$$

Which of the following choices hold?

[GATE-2007, 1-Mark]

IIT-K

(A) (i) is false, but (ii) and (iii) are true

☒ (B) (i) and (iii) are false, but (ii) is true

(C) (i) and (ii) are false, but (iii) is true

(D) (i), (ii) and (iii) are false

Ans: B

#Q. Suppose two hosts use a TCP connection to transfer a large file. Which of the following statements is/are False with respect to the TCP connection?

- ✓ I. If the sequence number of a segment is m , then the sequence number of the subsequent segment is always $m+1$. **FALSE**
- II. If the estimated round trip time at any given point of time is t sec, the value of the retransmission timeout is always set to greater than or equal to t sec. **TRUE**
- ✓ III. The size of the advertised window never changes during the course of the TCP connection. **FALSE**
- IV. The number of unacknowledged bytes at the sender is always less than or equal to the advertised window. **TRUE**

[GATE-2015, Set-1, 1-Mark]

(A) III only

(C) I and IV only

✓ (B) I and III only

(D) II and IV only

11T-K

Ans: B

[MSQ]

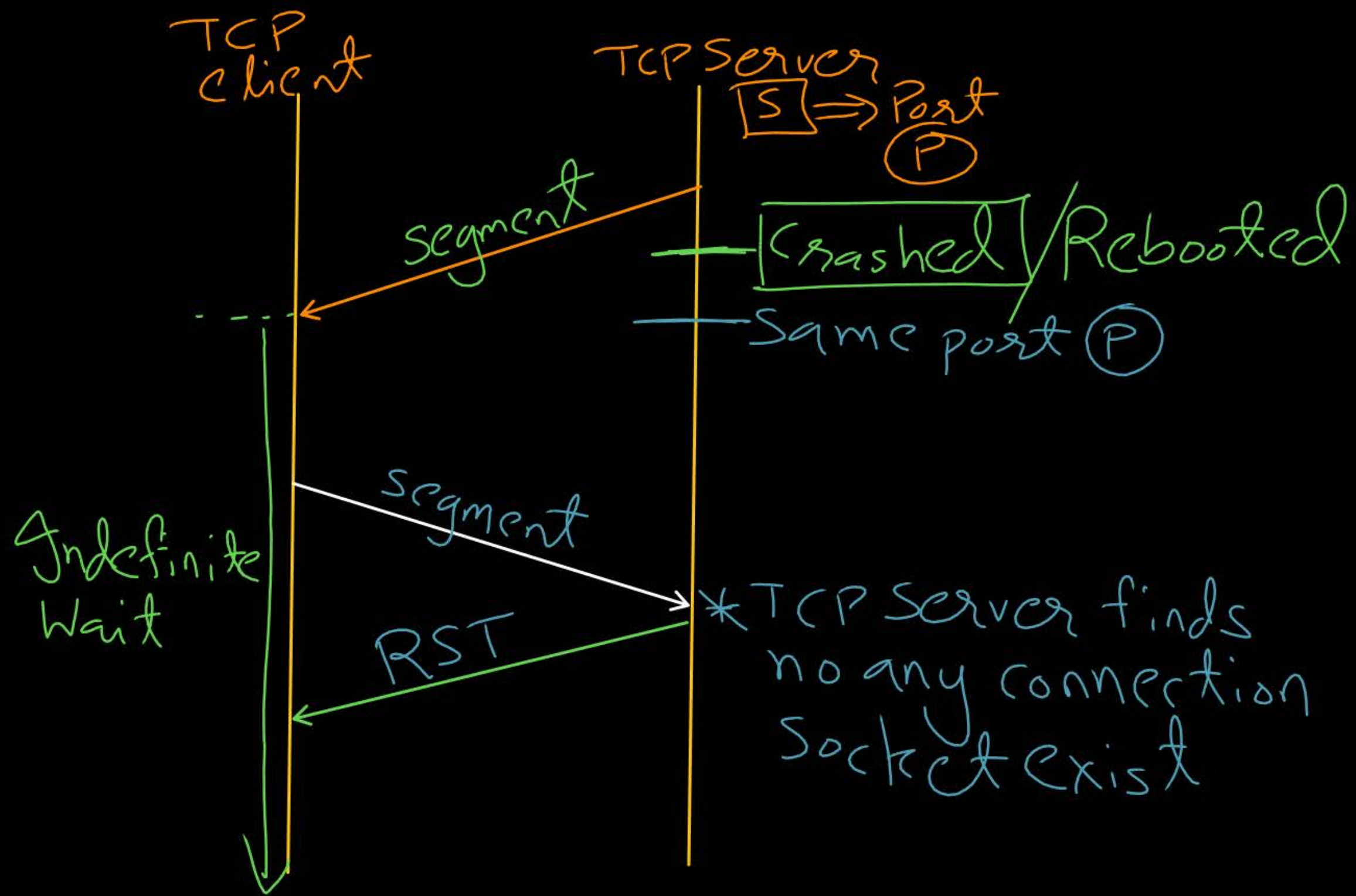
[IIT-B] [GATE-2021][2 Mark]



#Q. A TCP server application is programmed to listen on port number P on host S . A TCP client is connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviors is/are possible?

- ☒ A If the client was waiting to receive a packet, it may wait indefinitely TRUE
- ☒ B The TCP server application on S can listen on P after reboot → TRUE for some OS
- ☒ C If the client sends a packet after the server reboot, it will receive a RST segment
- ☒ D If the client sends a packet after the server reboot, it will receive a FIN segment

Ans: A, B & C



#Q. Which one of the following statements is FALSE?

[GATE-2004, 1-Mark]

Disadv. of TCP

- ✓ (A) TCP guarantees a minimum communication rate FALSE IIT-D
- ✗ (B) TCP ensures in-order delivery TRUE
- ✗ (C) TCP reacts to congestion by reducing sender window size TRUE
- ✗ (D) TCP employs retransmission to compensate for packet loss TRUE

Ans: A

#Q. Packets of the same session may be routed through different paths in :

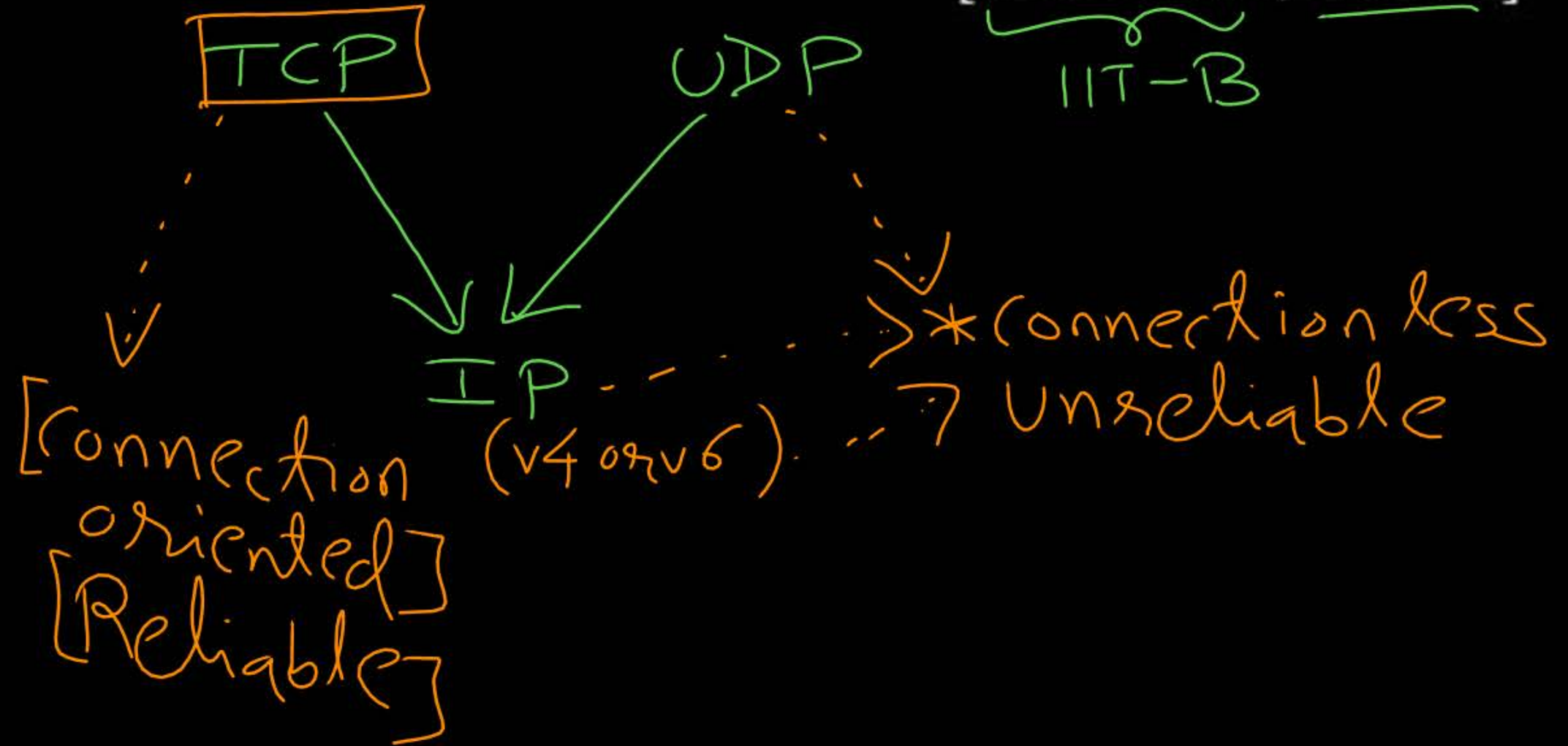
[GATE-2005, 1-Mark]
IIT-B

(A) TCP, but not UDP

✓ (B) TCP and UDP

(C) UDP, but not TCP

(D) Neither TCP, nor UDP



Ans: B



Topic : Urgent Pointer



- Urgent Pointer (16-bits)
- Valid only if URG (urgent) flag is set (on)
- Defines the number to be added to the [sequence number]
to determine the last urgent byte in the payload



Topic : Push Flag



- Do not buffer the data in 'Recv Buffer'
[signals the receiver to immediately forward the received data to the application]
- Push it immediately to the (application)
[bypassing the buffer and reducing latency]



Topic : Silly Window Syndrome (SWS)

- When TCP receiver advertises smaller value in 'Window Size'
- And TCP sender sends smaller size TCP segment
[as per advertised smaller value in 'Window Size']
- Lead more overhead in the network

TCP Sender

TCP Recv.



W.S. = 10

TCP seg.

Payload: 10 byte



Topic : TCP Optional Header

- Also known as “Options”
- Options are variable in size
- Maximum size of TCP optional header is (10 words) $\Rightarrow 10 \times 4 \text{ bytes}$
[Words of 32 bits, i.e. 40 bytes]
- if options size are not in words
than it uses padding [End-of-Option Option]



Topic : TCP Optional Header

→ Options field component :-

1, Option Kind (1 byte) ✓

2. Option Length (1 byte) }

3. Option Data



Topic : TCP Optional Header

BASE Header
[5 Words] $\Rightarrow$ 20 bytes

Option Kind	option length	MSS
option kind	option length	shift-count

(Options)

Payload



Topic : TCP Options



Kind	Length	Purpose
0	-	End of Option [Padding]
1	-	No Operation
2	4 bytes	Maximum Segment Size ✓
3	3	Window Scale
4	2	Selective ACK Permitted
5	10 / 18 / 26 / 34	Selective ACKs (SACK)
8	10	Timestamp



2 mins Summary



Topic

TCP Options



THANK - YOU

